

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 5

2 Computer Networking (awm22)

A networking enthusiast says, “TCP does not perform well for the very long, the very short, the very fast, or the very slow.”

- (a) When and why are they correct? Explain your answer clearly. [8 marks]

Answer:

The statement allows the candidate to ignore the nature of packet loss, this is a question about using a network efficiently: adapting networks to the available bandwidth.

Subsequently I'd expect the candidate to discuss the bandwidth delay product - in particular how the operating regime of TCP means that the BDSP is limited in extremes; extremes expressed as "very fast/slow" and "very long/short";

- (b) QUIC is a new transport protocol that offers many advantages over TCP, it may not solve the issues the enthusiast of the last question discusses. Prototyping new protocols to replace TCP has proven difficult in the past. Discuss the QUIC approach to this particular challenge. [2 marks]

Answer:

several parts for this answer these two are critical

- UDP
- user space implementation

Clearly the acronym for QUIC can't be put here as it offers significant help.

- (c) Considering the capacity and limits of end-system components (e.g., CPU(s), cache(s), main memory, peripheral interconnects, network adapters), propose a practical strategy for end-systems to deliver the next generation of Ethernet speeds (100Gbit/s and 400Gbit/s).

Alongside your strategy you may wish to also consider the particular challenges invoked by improved security in networking such as the wider use of IP header encryption (IPsec), the increasing use of full application-payload encryption, and so-forth.

Justify your approach by stating your assumptions throughout.

Hint: your approach need not be universal. It may help to consider aspects of Computer Architecture, Algorithms and Computer Networking in your approach. [10 marks]

Answer:

The hint is the clue:

pick the target objective.

transport
protocols reliable
datagram

— *Solution notes* —

if simply vomiting packets or capturing packets, then time-keeping is more important
Systems that sink and source 400 or 100Gbps are so special as to be in a 0.001% of hosts problem

The challenge of crypto is increasing assumptions about it being an end-app-level process which fundamentally limits any encrypted flow to the speed of communication between two cores on two different machines (maximum is dictated by NIC location in host, which socket? cores sharing cache? DPDK type approaches (eating cores for hierarchies of steering), trading BW for latency in the host via such things as batching, aggregation, IRQ rates and so forth.

Current NIC tech is 10Gbps without some offloading (per-packet offload), 40Gbps without some steering and stream-splitting - a single core on a single CPU simple is unable to process (raw) IP packets at a high enough rate.

The solution is app-offloading but the challenge is distributed state management.
